Subarray Sums II

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1661 | Sorting and Searching | https://cses.fi/problemset/task/1661

Given an array of n integers, your task is to count the number of subarrays having sum x.

Input

The first input line has two integers n and x: the size of the array and the target sum x.

The next line has n integers $a_1,a_2,\ldots,a_n$: the contents of the array.

Output

Print one integer: the required number of subarrays.

Constraints

- $1 \le n \le 2 \cdot 10^5$
- $-10^9 \le x,a_i \le 10^9$

Example

Input
5 7 2 -1 3 5 -2

Output
2